Magical Potion Brewing

Problem Description

Ilya works in a magic laboratory. Today, she has received a boring task: brewing a magical potion. The laboratory initially has only one raw material numbered 1. Ilya needs to use m known brewing reactions to step-by-step transform this raw material into a magical potion numbered n.
Brewing each new material takes a corresponding time a_i. However, some materials are too active, and brewing them causes irreversible damage to the cauldron, making all subsequent brewing processes take an additional time b_i (except including the current brewing process).
Since the brewing process is very tedious, Ilya hopes you can help her design a synthesis route that minimizes the total time cost.

Input and Output

Input Format

The input consists of $m + 3$ lines.
Line 1 contains 2 integers: n and m. Here, n is the number of known magical materials, and m is the number of known brewing reactions.
Line 2 contains n positive integers, $a_1, a_2, \cdots, a_n$, representing the time required to brew each magical material.
Line 3 contains n positive integers, $b_1, b_2, \cdots, b_n$, representing the additional time caused by brewing each magical material for all subsequent brewing processes.
The next m lines each contain 2 integers, u and v, indicating that the magical materials numbered u and v can be transformed into each other through brewing reactions.

Output Format

Output an integer representing the minimum total time cost to brew the magical potion.

Data Range

- For 30% of the data, $n \leq 500$, $\sum_{i=1}^{n} b_i \leq 1000$;
- For 60% of the data, $n \leq 2000$, $\sum_{i=1}^{n} b_i \leq 5000$;

- For 90% of the data, $n \le 5000$, $\sum_{i=1}^{n} b_i \le 10000$;
- For 100% of the data, $n \le 10000$, $\sum_{i=1}^{n} b_i \le 25000$;
- For all data, $0 < a_i \le 10$, $a_1 = 0$, $0 \le b_i \le 100$, $b_1 = 0$, $m \le \frac{n^2-n}{2}$;

Sample Data

Sample Input 1

```
6 6
0 2 5 4 3 1
0 5 0 0 0 1
1 2
1 3
2 5
3 4
4 5
5 6
```

Sample Output 1

```
13
```

Sample Input 2

```
6 6
0 9 1 2 2 3
0 0 1 1 0 0
1 2
1 3
2 5
3 4
4 5
5 6
```

Sample Output 2

```
13
```

Sample Input 3

```
9 10
0 8 9 3 5 1 2 2 1
0 0 0 1 2 3 2 0 0
1 2
1 4
1 6
2 3
3 8
4 5
5 8
6 7
7 8
8 9
```

Sample Output 3

```
18
```